

19-200-0105B

Basic Mechanical Engineering

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Division of Mechanical Engineering

Syllabus

19-200-0105B/19-200-0205A BASIC MECHANICAL ENGINEERING

Course Outcomes:

On completion of this course the student will be able to:

1. Understand basics of thermodynamics and working of steam turbines
2. Understand basics of internal combustion engines, refrigeration and air conditioning
3. Gain knowledge on the working of hydraulic turbines and centrifugal pumps
4. Identify manufacturing methods encountered in engineering practice and understand mechanism of power transmission

Module I

Thermodynamics: Thermodynamics systems – open, closed and isolated systems, equilibrium state of a system, property and state, process, cycle, Zeroth law of thermodynamics- concept of temperature, temperature scales. First law – internal energy, enthalpy, work and heat, Different processes (isobaric, isochoric, isothermal, adiabatic and polytropic processes). Second law – Kelvin-plank and Clausius statements and their equivalence, Carnot Cycle (Elementary problems only). Thermodynamic properties of Steam, Steam Generator. Different types of boilers, boiler mountings and accessories. Formation of steam at constant pressure, working of steam turbines, compounding of turbines.

Module II

Internal Combustion Engines: Air standard cycles – Otto and Diesel cycles, working of two stroke and four stroke Petrol and Diesel engines, Carburetted and MPFI engines, fuel pump, fuel injector, ignition system, cooling system, lubricating system.

Refrigeration & Air-conditioning: Introduction to refrigeration and air -conditioning, Rating of refrigeration machines, Coefficient of performance, Simple refrigeration vapour compression cycle (Elementary problems only), Summer and winter air conditioning.

Module III

Hydraulic Turbines & Pumps: Introduction, Classification, Construction details and working of Pelton Wheel, Francis and Kaplan turbines, Specific speed (Definition and significance only),



Classification of water pumps, working of centrifugal pumps and reciprocating pumps (Theory of working principles only)

Power plants: Hydro-electric power plants, Thermal power plants, Nuclear power plants, Diesel power plants, Wind mills, solar energy (Working principles using schematic representations only)

Module IV

Introduction to Manufacturing Systems: Welding- different types of welding, resistance welding, arc welding, gas welding, Brazing and soldering, Different welding defects. Casting- different casting processes, sand casting, casting defects, Rolling- hot rolling and cold rolling, two high, three high, cluster rolling mills, wire drawing, forging, extrusion, Heat treatment of steel, elementary ideas of annealing, hardening, normalizing, surface hardening.

Power Transmission Methods and Devices: Introduction to Power transmission, Belt, Rope, Chain and Gear drive. Length of belt open and crossed. Ratio of belt tensions (Elementary problems only). Different types of gears (Elementary ideas only). Types and functioning of clutches.

References

1. Nag, P.K. Engineering thermodynamics. (Fifth Edition). McGraw Hill Education (India) Pvt. Ltd, New Delhi.(2013).
2. Gill, J.H. Smith Jr. and Ziurys, E.J. Fundamentals of internal combustion engines, Oxford & IBH, New Delhi.(1959)
3. Stoecker, W. F. Refrigeration and air conditioning. Tata McGraw Hill, New Delhi. (1980).
4. JagadishLal. Hydraulic machines. Metropolitan Book co, New Delhi.(1994)
5. Raghavan, V. Material science and engineering, Prentice Hall of India, New Delhi. (2004)
6. Rajendar Singh.Introduction to basic manufacturing processes and workshop technology, New Age International, New Delhi. (2006).

Module-2a: Internal Combustion Engines



Gas Power Cycle



- Working fluid:- Gas/Air
- Examples:- Otto cycle, Diesel cycle, Gas Turbine cycle etc.
- Energy is provided by burning fuel within the boundary
- Air + Fuel => Combustion products
- But air is predominantly Nitrogen, it hardly undergoes any chemical reaction during combustion, working fluid closely resembles air

Air-Standard Cycle



- Approximations used for analysis of Gas Power Cycle (Air-standard cycle)
- Working fluid is Air, which continuously circulates in a closed loop and always behaves like an ideal gas
- All processes are internally reversible
- Combustion process is replaced by heat addition process from an external source
- Exhaust process is replaced by heat rejection process

Overview of Reciprocating Engine

- Top Dead Center (TDC):- Smallest volume
- Bottom Dead Center (BDC):- Largest volume
- Bore
- Stroke
- Inlet and outlet valves
- Clearance volume:- Min vol

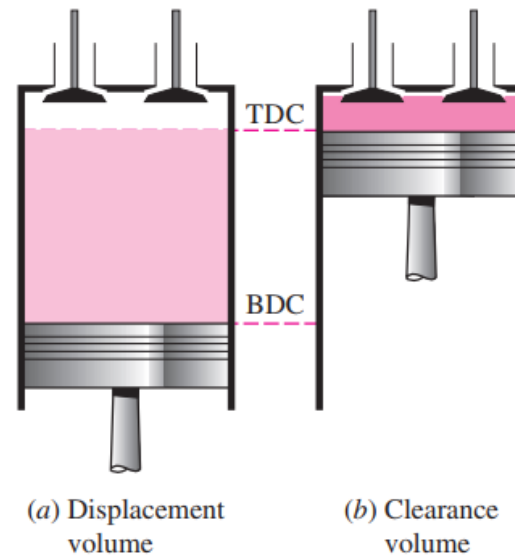


FIGURE 9-11

Displacement and clearance volumes of a reciprocating engine.

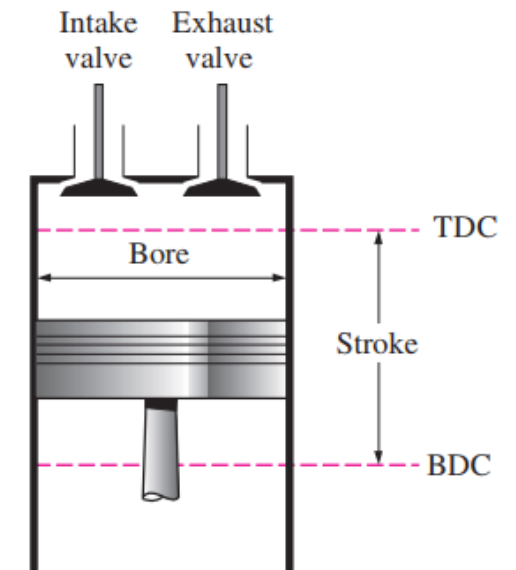


FIGURE 9-10

Nomenclature for reciprocating engines.

Overview of Reciprocating Engine

- Displacement volume:- volume displaced by piston as it moves from TDC to BDC
- Compression Ratio = Ratio of max. volume to min volume

$$r = \frac{V_{\max}}{V_{\min}} = \frac{V_{\text{BDC}}}{V_{\text{TDC}}}$$

- Reciprocating Engines (based on the way combustion initiated)
 - Spark Ignition Engine
 - Compression Ignition Engine

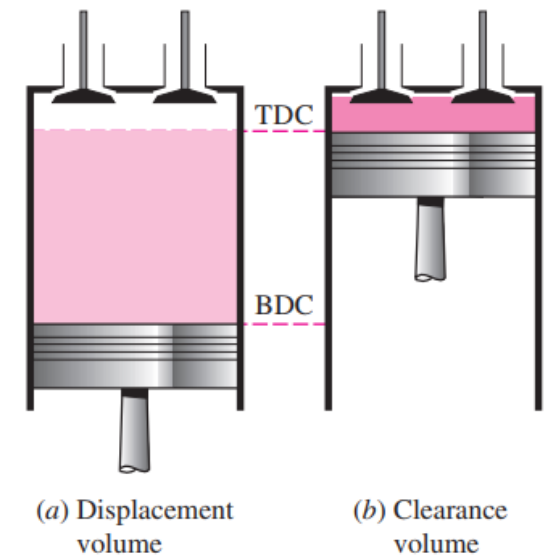
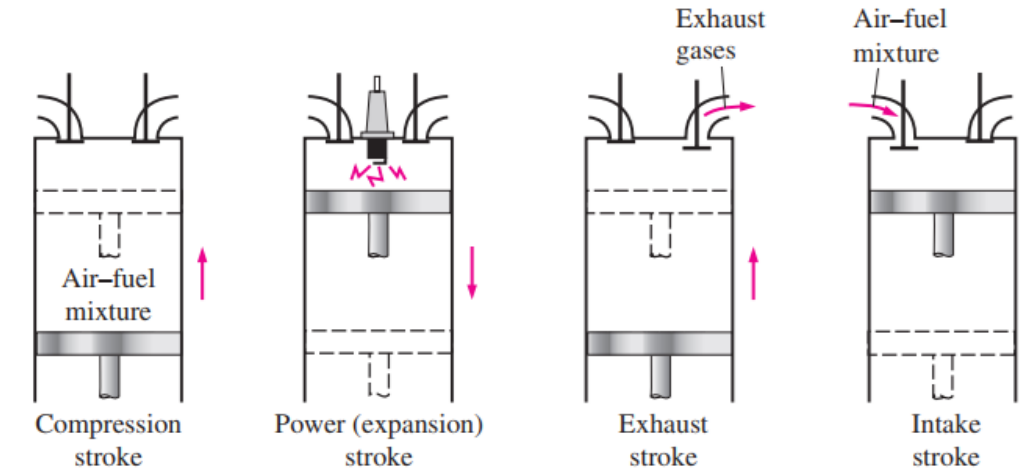
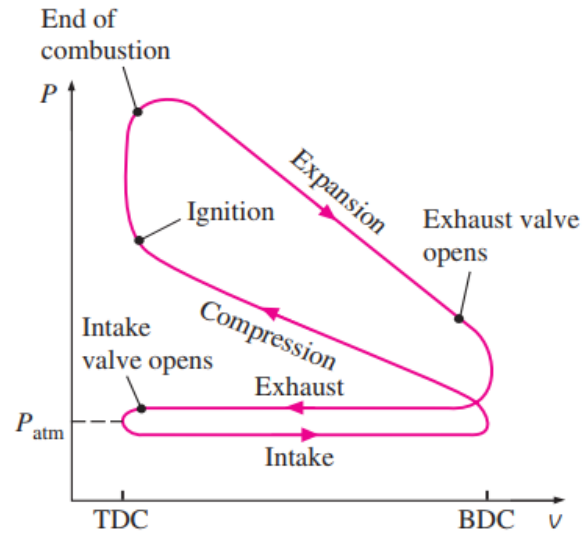


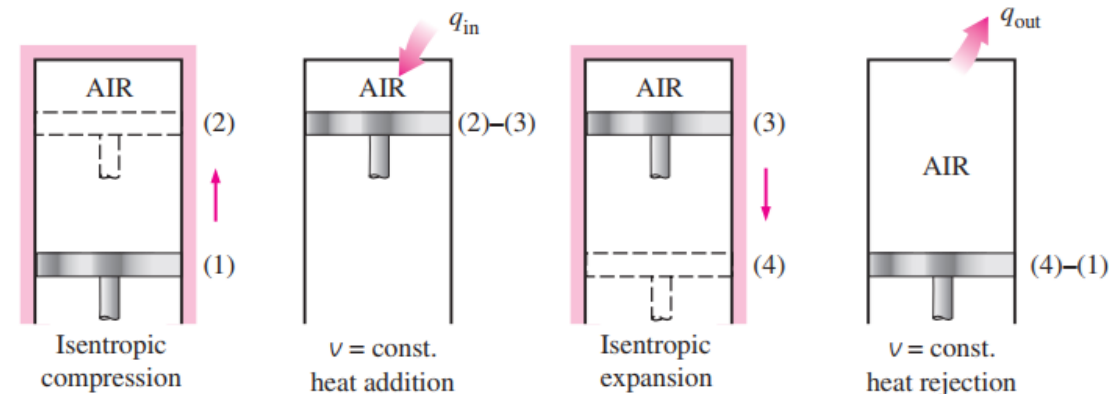
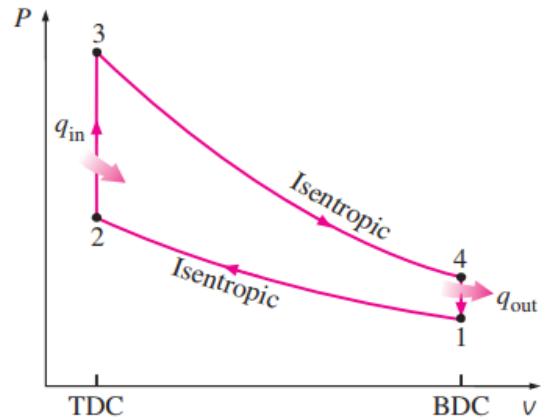
FIGURE 9-11

Displacement and clearance volumes of a reciprocating engine.

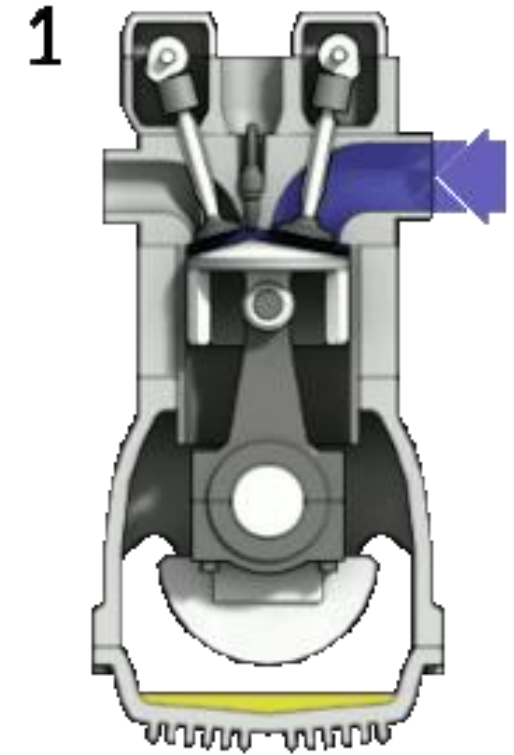
Otto Cycle:- Ideal cycle for Spark Ignition Engines



(a) Actual four-stroke spark-ignition engine



(b) Ideal Otto cycle



Otto Cycle:- Analysis

1-2 Isentropic compression

2-3 Constant-volume heat addition

3-4 Isentropic expansion

4-1 Constant-volume heat rejection

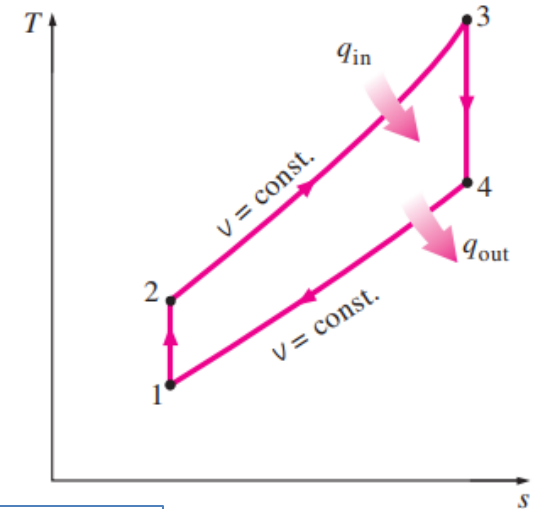
$$(q_{in} - q_{out}) + (w_{in} - w_{out}) = \Delta u \quad (\text{kJ/kg})$$

$$q_{in} = u_3 - u_2 = c_v(T_3 - T_2)$$

$$q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$$

$$\eta_{th,Otto} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)}$$

$$\frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{k-1} = \left(\frac{v_3}{v_4}\right)^{k-1} = \frac{T_4}{T_3}$$



$$\eta_{th,Otto} = 1 - \frac{1}{r^{k-1}}$$

$$r = \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = \frac{v_1}{v_2}$$



Problem-I



A vehicle powered by a spark-ignition engine follows air standard Otto cycle ($\gamma = 1.4$). The engine generates 70 kW while consuming 10.3 kg/hr of fuel. The calorific value of fuel is 44,000 kJ/kg. The compression ratio is _____ (correct to two decimal places).

Concept:

A spark-ignition engine works upon Otto cycle

Air standard, efficiency for an Otto cycle is given by $\eta = 1 - \frac{1}{r^{\gamma-1}}$

Where 'r' is compression ratio, $\gamma = 1.4$ for air

Efficiency is also given by

$$\eta = \frac{\text{Power output}}{\text{Heat input}}$$

Heat input = Q = (mass flow rate of fuel) $\times$ (calorific value of fuel) = $\dot{m} \times cv$

Calculation:

Given, $\dot{m} = 10.3 \text{ kg/hr}$, $cv = 44,000 \text{ kJ/kg}$, Power output = 70 kW

$$Q = \left(\frac{10.3}{3600} \times 44,000 \right) kW = 125.89 \text{ KW}$$

$$\eta = \frac{70}{125.89} = 0.556$$

Now,

$$0.556 = 1 - \frac{1}{r^{(1.4-1)}} \Rightarrow \frac{1}{r^{0.4}} = 0.444$$

$$\Rightarrow r = 7.61$$

Diesel Cycle

- Working very similar to SI Engine, only difference is: method of initiating combustion
- SI Engine:- Air-Fuel mixture is compressed below autoignition temperature, and combustion is initiated with the help of spark plug
- CI Engine:- Air is compressed to temp above auto ignition temp, and fuel is injected into compressed air
- Diesel Engine compression ratio:- 14-24

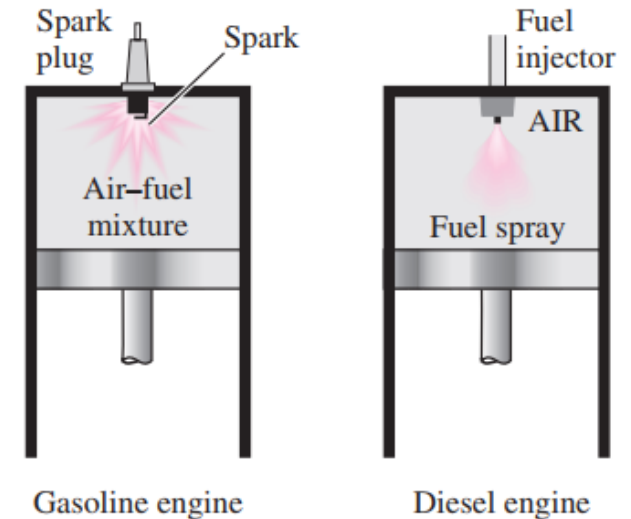
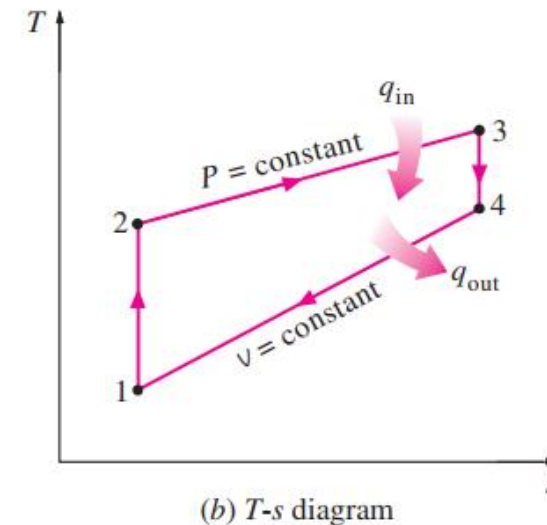
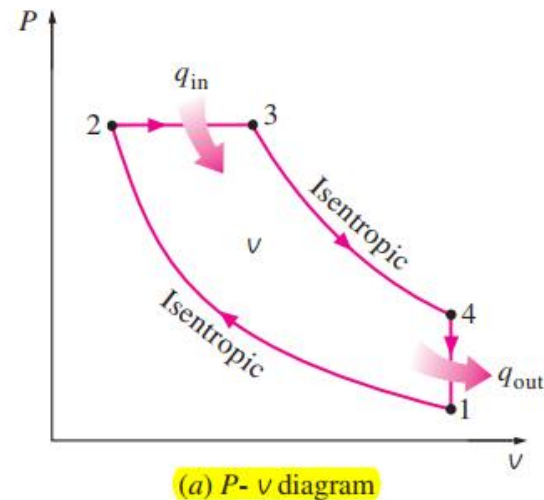


FIGURE 9-20

In diesel engines, the spark plug is replaced by a fuel injector, and only air is compressed during the compression process.

Diesel Cycle

- The fuel injection process in diesel engines starts when the piston approaches TDC and continues during the first part of the power stroke.
- The combustion process in these engines takes place over a longer interval.



Diesel Cycle

$$\begin{aligned}q_{\text{in}} - w_{b,\text{out}} &= u_3 - u_2 \rightarrow q_{\text{in}} = P_2(v_3 - v_2) + (u_3 - u_2) \\&= h_3 - h_2 = c_p(T_3 - T_2)\end{aligned}$$

$$-q_{\text{out}} = u_1 - u_4 \rightarrow q_{\text{out}} = u_4 - u_1 = c_v(T_4 - T_1)$$

$$r_c = \frac{V_3}{V_2} = \frac{v_3}{v_2}$$

$$\eta_{\text{th,Diesel}} = \frac{w_{\text{net}}}{q_{\text{in}}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{T_4 - T_1}{k(T_3 - T_2)} = 1 - \frac{T_1(T_4/T_1 - 1)}{kT_2(T_3/T_2 - 1)}$$

$$\eta_{\text{th,Diesel}} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$$

- For same compression ratio:

$$\eta_{\text{th,Otto}} > \eta_{\text{th,Diesel}}$$



Problem-2

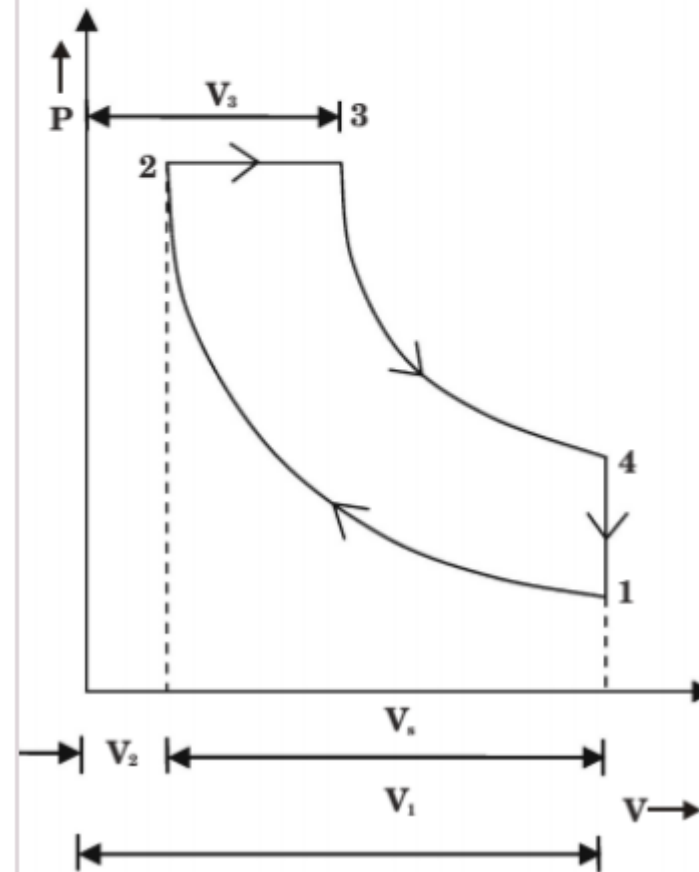
. A diesel engine has a compression ratio of 17 and cut-off takes place at 10% of the stroke. Assuming ratio of specific heats (γ) and the air standard efficiency (in%) is –.

Compression ratio

$$= 17 = \frac{V_1}{V_2} = V_1 = 17V_2$$

where ρ = cut-off ratio

$$= \frac{V_3}{V_2}$$



$$V_3 - V_2 = \frac{10}{100} V_s$$

$$V_3 - V_2 = 0.1(V_1 - V_2)$$

$$= 0.1(17V_2 - V_2)$$

$$= 0.1(16V_2)$$

$$V_3 = 1.6V_2 + V_2$$

$$V_3 = 2.6V_2$$

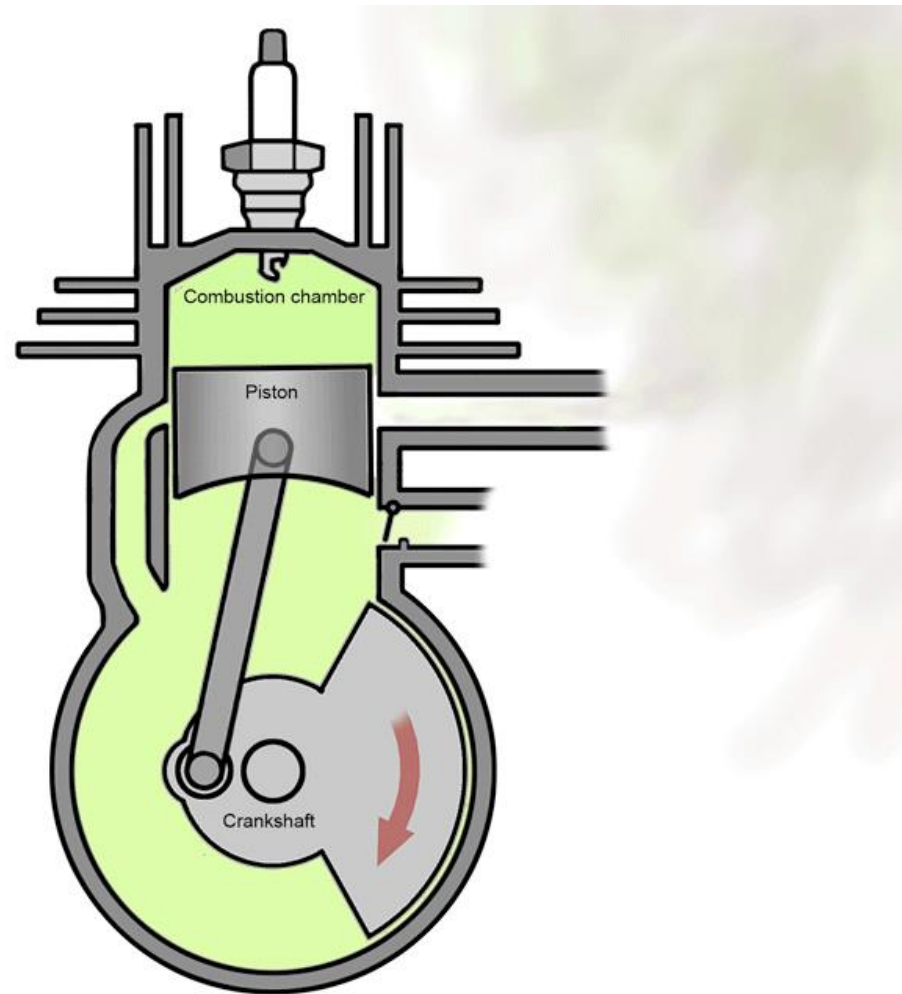
$$= \frac{V_3}{V_2} = \rho = 2.6$$

$$\eta = 1 - \frac{1}{\gamma(r)^{\gamma-1}} \times \left[\frac{\rho^{\gamma} - 1}{\rho - 1} \right]$$

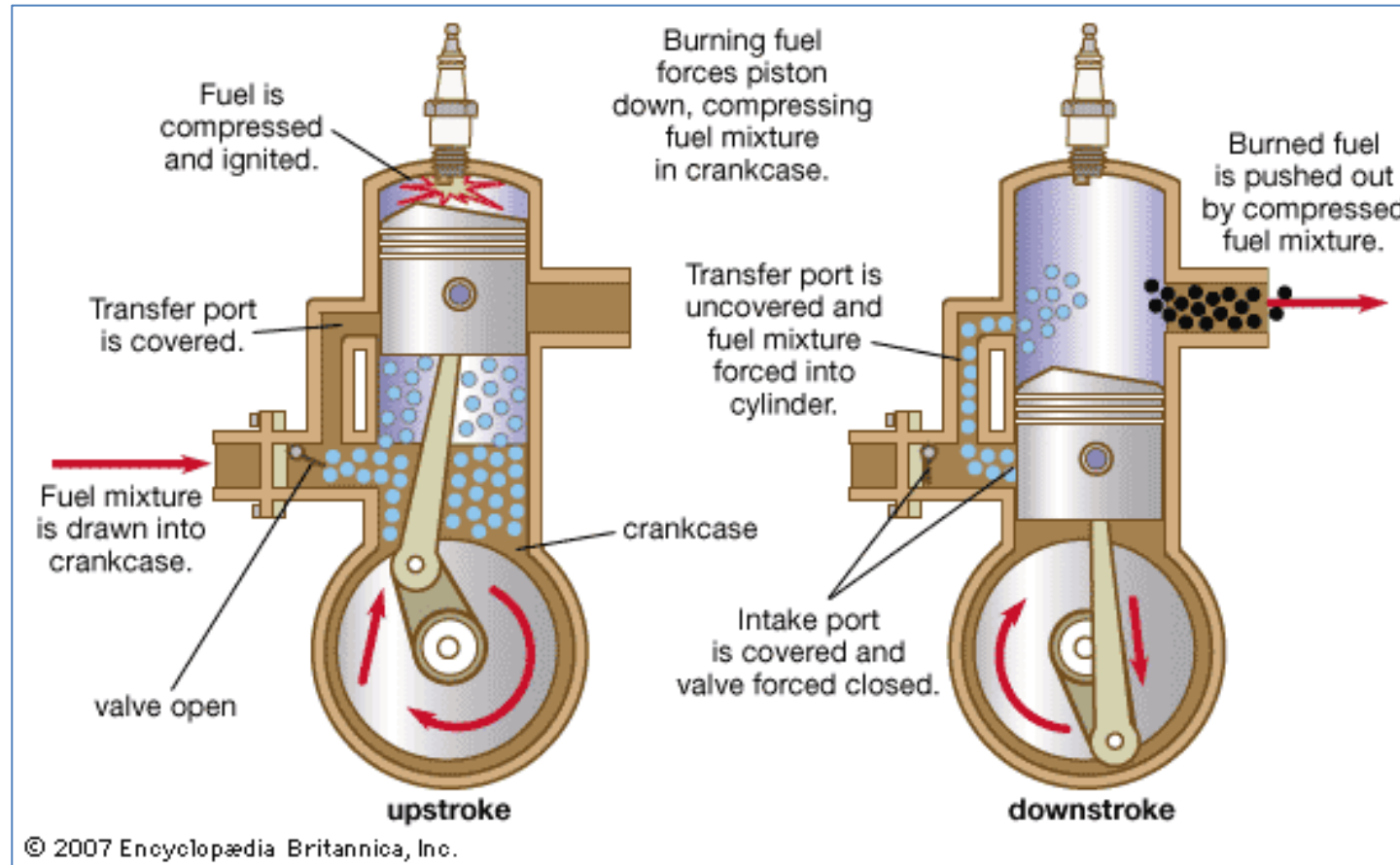
$$= 1 - \frac{1}{1.4(17)^{1.4-1}} \times \left[\frac{2.6^{1.4} - 1}{2.6 - 1} \right]$$

$$= 0.5960 = 59.60\%.$$

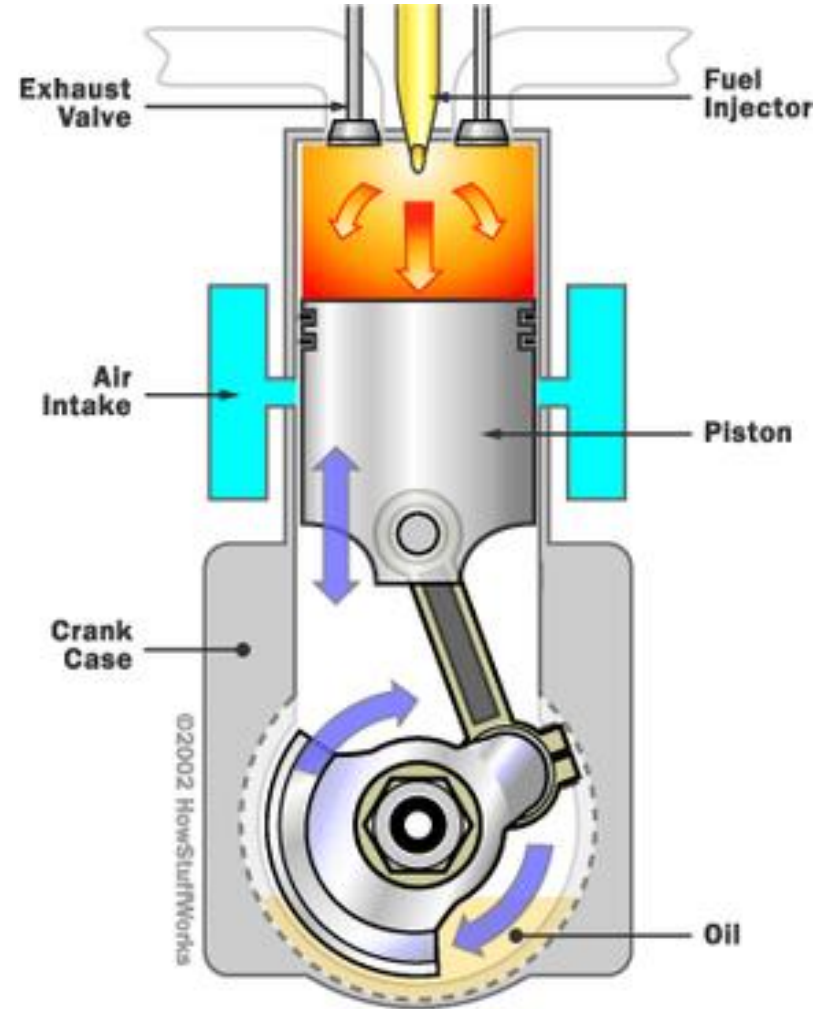
Two Stroke Engines



Two stroke Petrol Engine



Two stroke Diesel Engine



[2 Stroke Diesel Engine Animation - YouTube](#)

2 Stroke Vs 4 Stroke

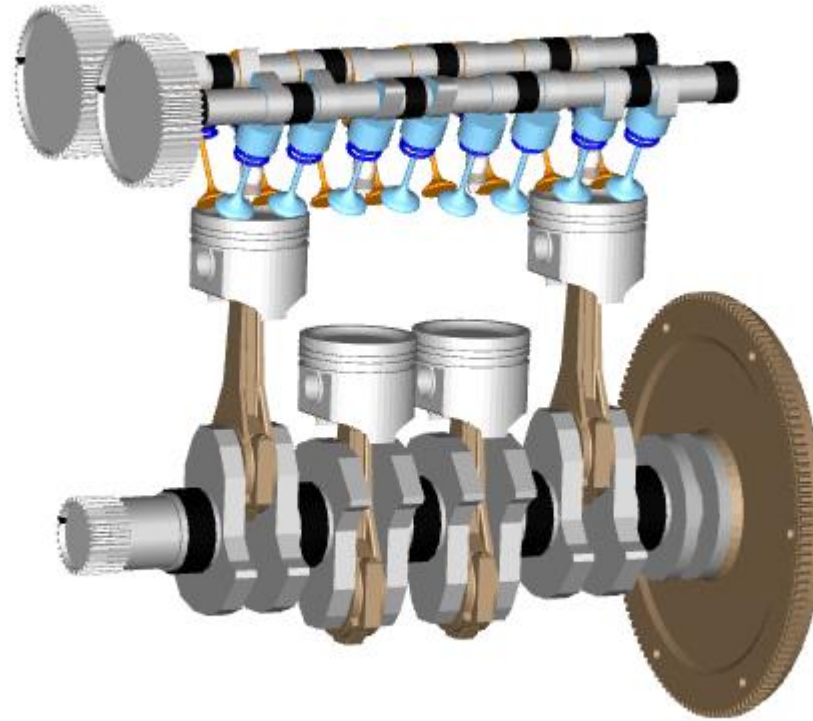
2Stroke

- One revolution of the crankshaft during one power stroke.
- Uses a port for fuel's outlet and inlet
- Lesser thermal efficiency
- Large ratio in terms of power to weights
- Requires more lubricating oil as some oil burns with the fuel
- More wear and tear due to poor lubrication
- Engines are lighter and noisy

4Stroke

- Two revolutions of the crankshaft during one power stroke
- Uses valves for fuel's outlet and inlet
- Higher thermal efficiency
- Lesser ratio in terms of power to weight
- Requires less lubricating oil
- Less wear and tear
- Engines are heavier because their flywheel is heavy and less noisy

Multi Cylinder Engines



Examples

- Single cylinder



- 2 Cylinder



Examples

- 3 Cylinders



- 4 Cylinders



Examples

- >6 Cylinder



Other applications



Engine Support Systems



1. Cooling system
2. Lubrication system
3. Fuel pump
4. Fuel injection
5. Ignition system
6. Carburetted and MPFI engine

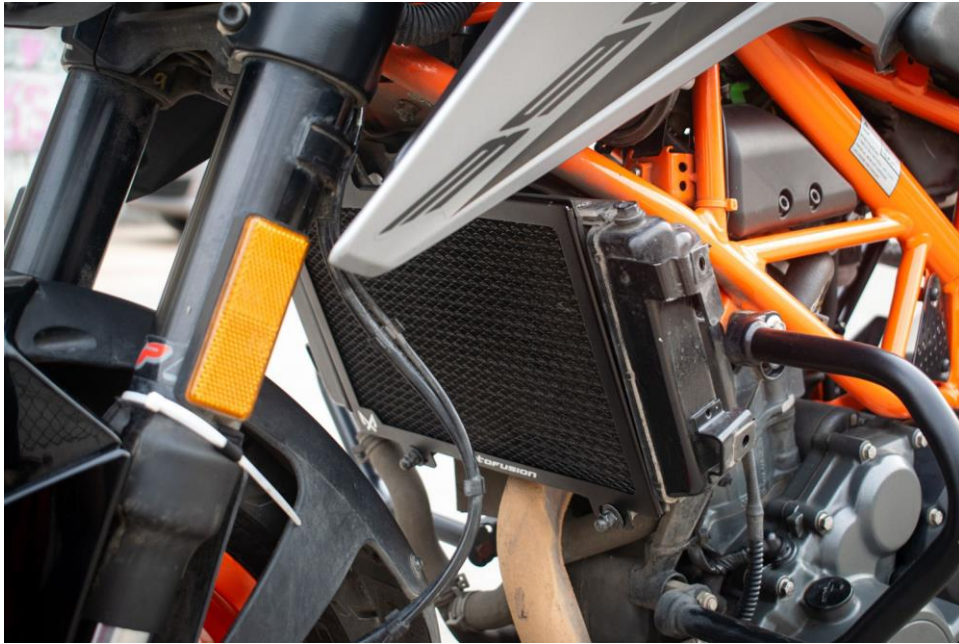
I. Cooling System



I. Engine Cooling System

- System control engine temperature
- If the heat generated in cylinder is not dissipated, it would result in the failure of the cylinder material
- Due to very high temperatures, the film of the lubricating oil will get oxidized, thus producing carbon deposits on the surface. This will result in piston seizure
- Thermal stresses in engine components
- Type of systems:
 1. Air cooling
 2. Liquid cooling

Engine Cooling System



Ia. Air Cooling

- Heat is conducted to the outer parts of the engine, is radiated and conducted away by the stream of air
- Fins (metallic ridges) are made to increase heat transfer area
- **Advantages:-**
 - Light weight
 - Need less care and maintenance
 - Cheaper to manufacture
- **Applications**
 - Motor cycles, scooters etc.

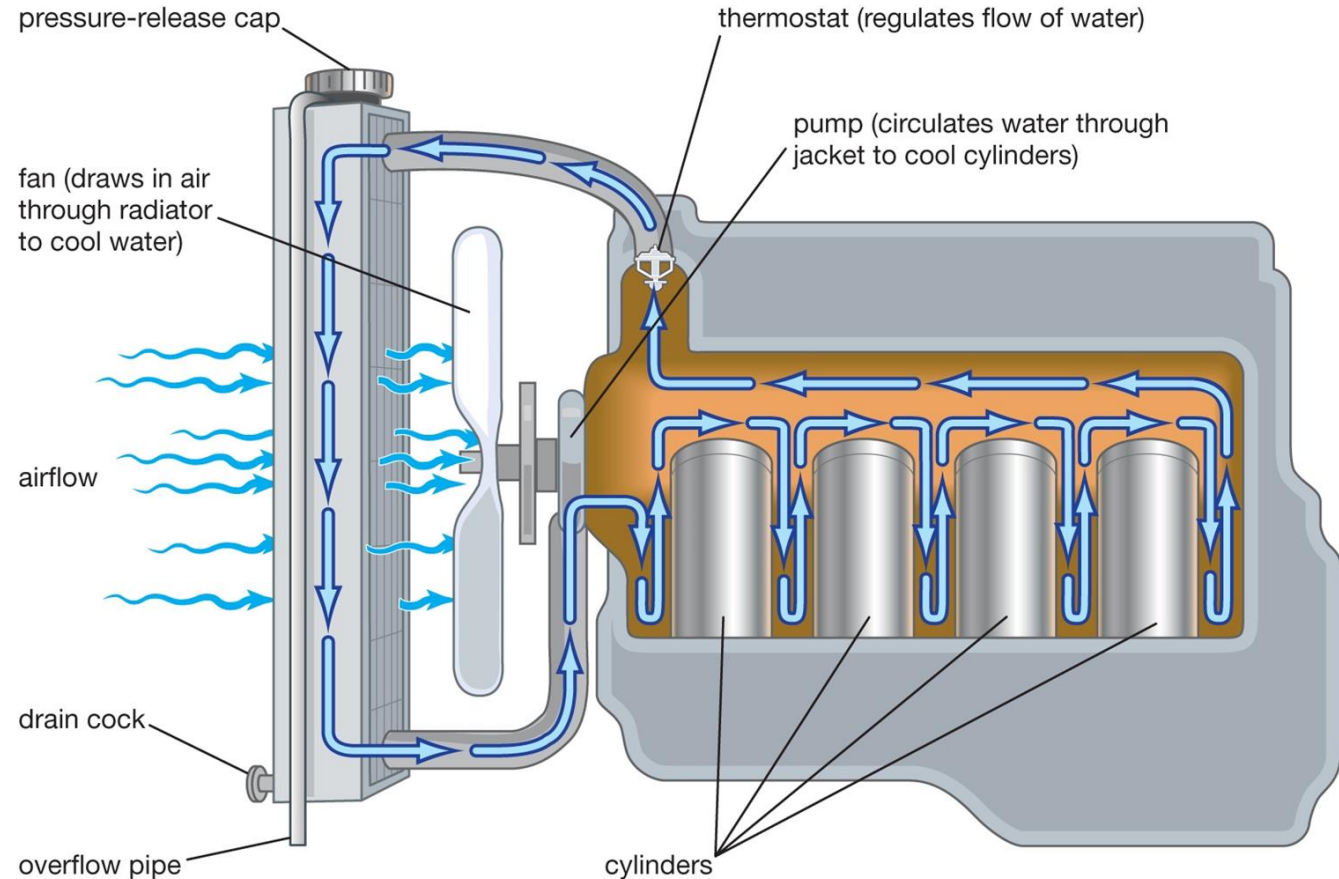


Ib. Liquid Cooling

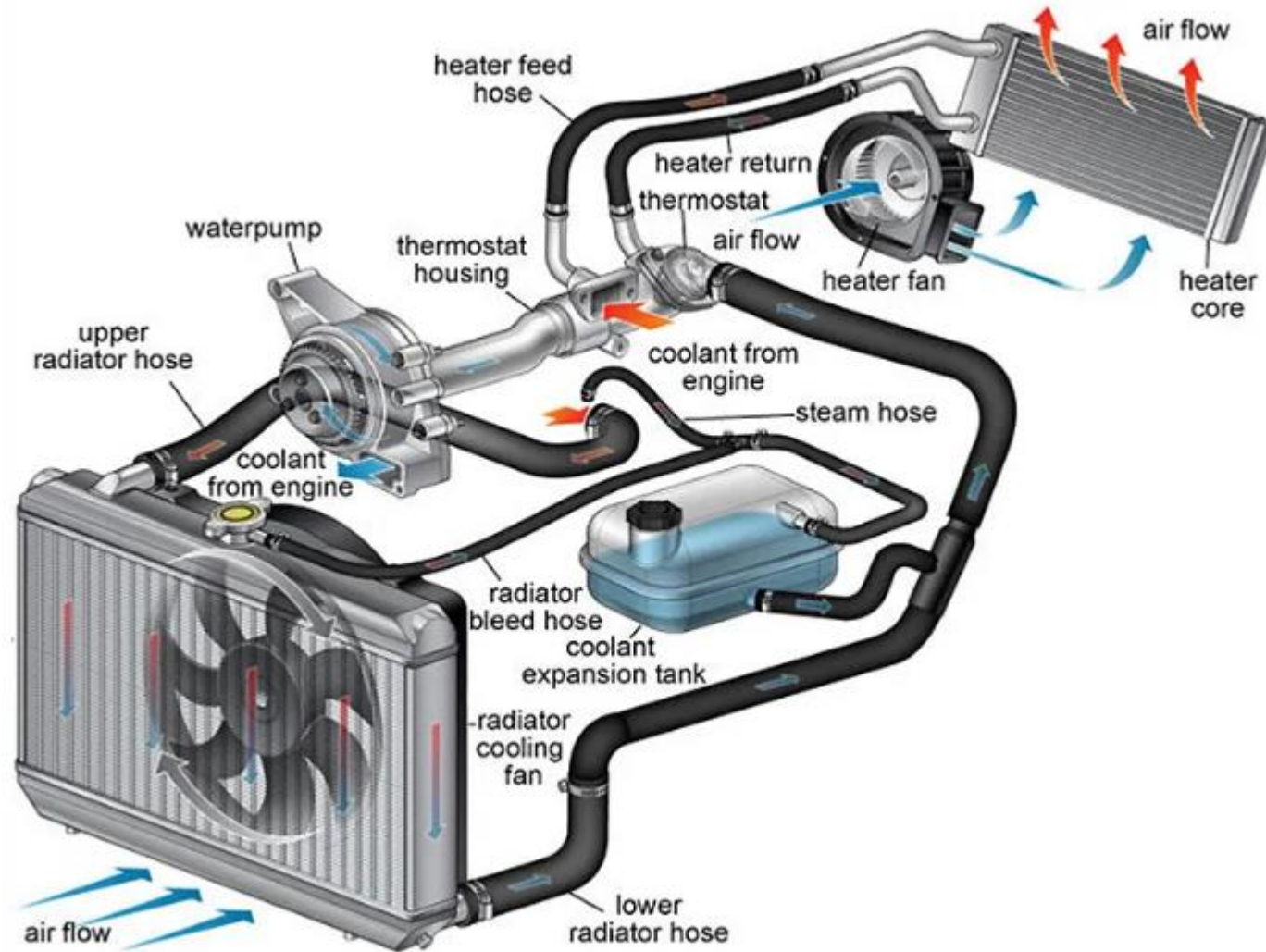
- Liquid as working fluid

Components

- A. water pump,
- B. fan
- C. radiator
- D. fan belt
- E. water jacket,
- F. thermostat valve,
- G. temperature gauge
- H. hose pipes



Cooling system



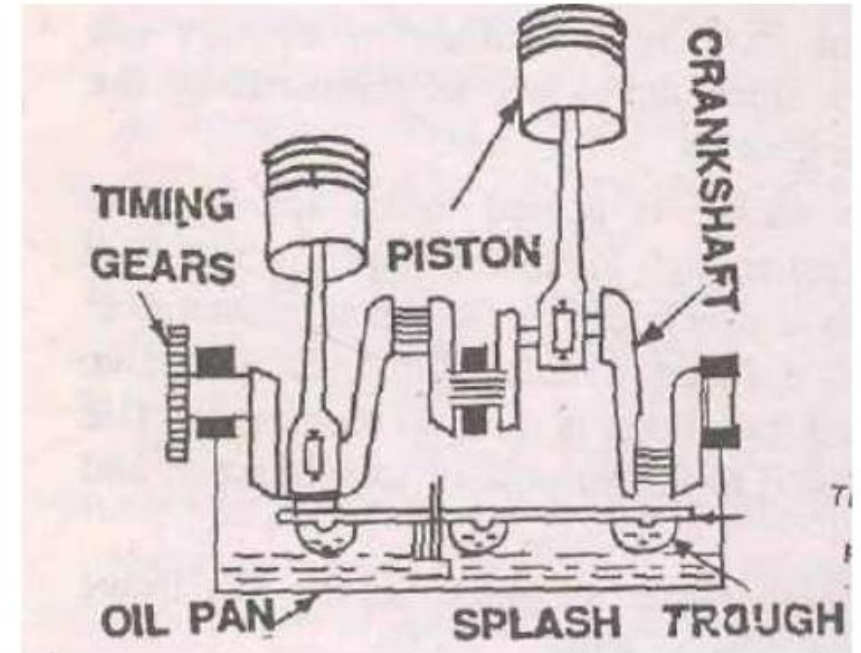
2. Engine Lubrication System

- Purpose:
 - Reducing friction effect
 - Cooling effect
 - Sealing effect
 - Cleaning effect
- A good lubricant should have the following qualities:
 - It should have sufficient viscosity to keep the rubbing surfaces apart
 - It should remain stable under changing temperatures.
 - It should keep lubricated parts clean.
 - It should not corrode metallic surfaces

Lubrication system

1. Splash lubrication system

- Oil is maintained at uniform level at oil pan
- Dipper dips into oil and splashes oil out of the pan
- Flashing action maintains mist of oil that drenches inner part of the engines
- Disadvantage:- non uniform lubrication

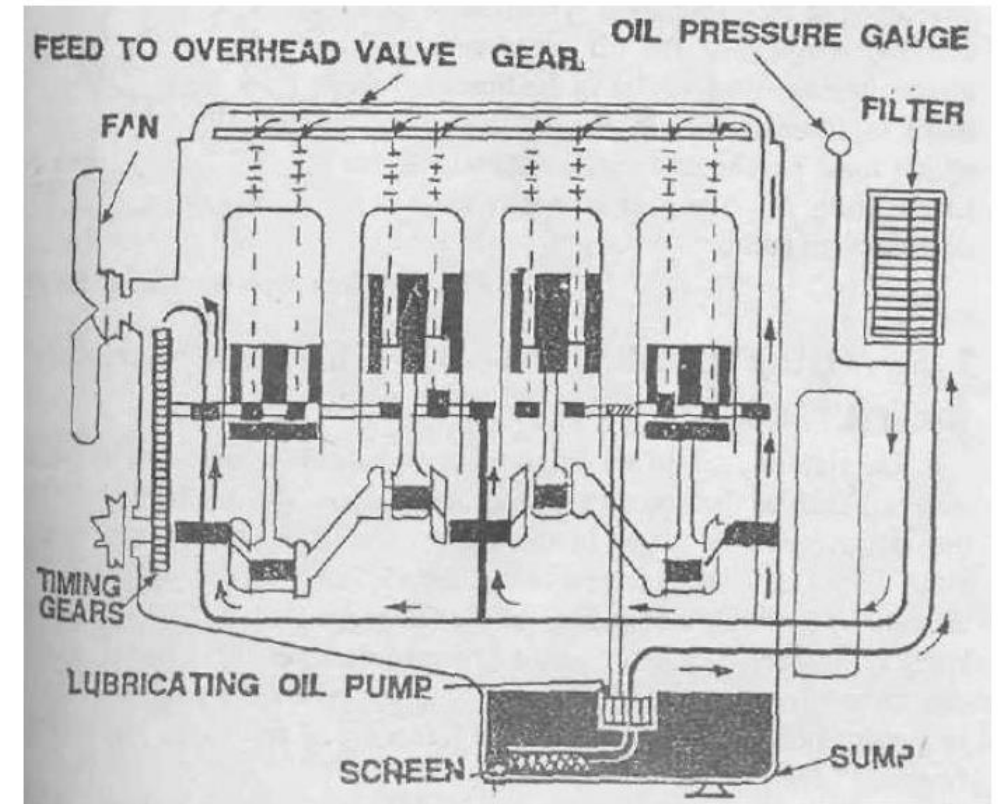


Splash lubrication system

Lubrication System

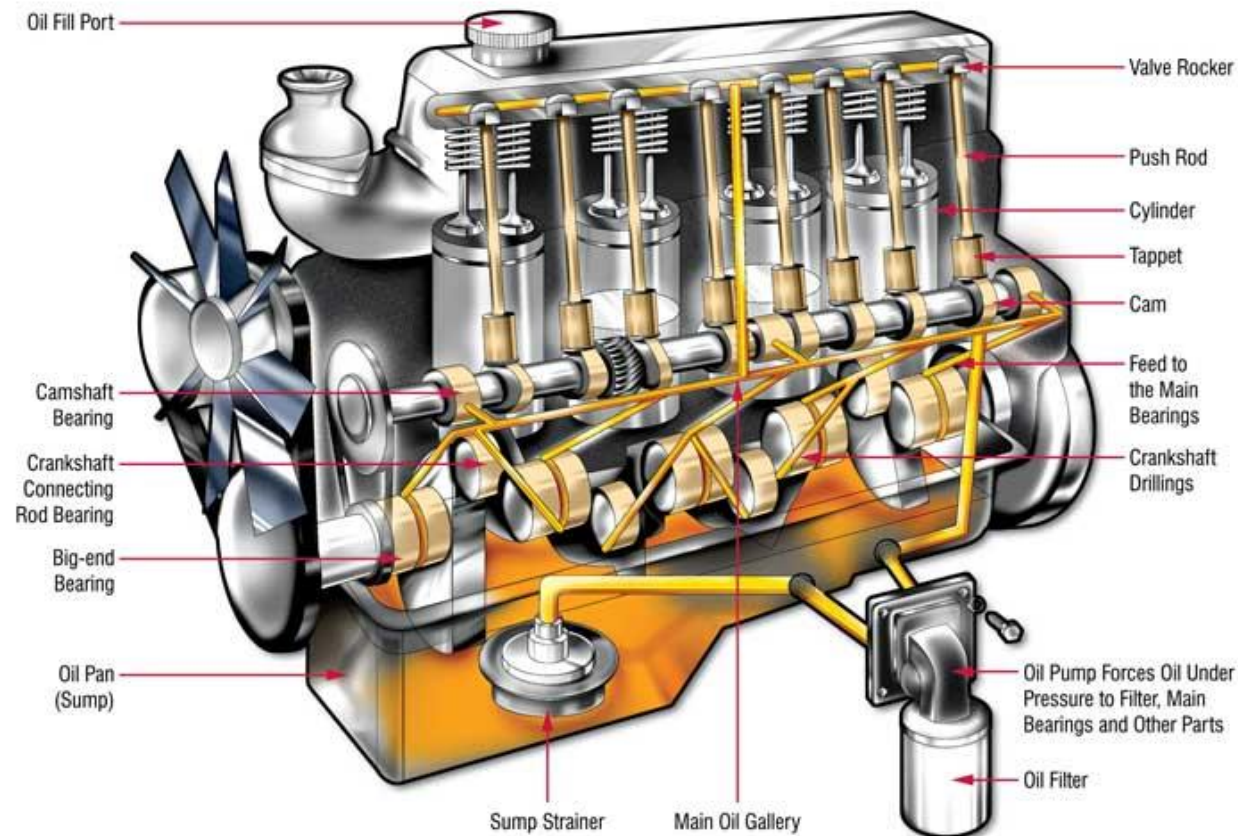
2. Forced feed lubrication system

- Oil is pumped directly to various engine parts



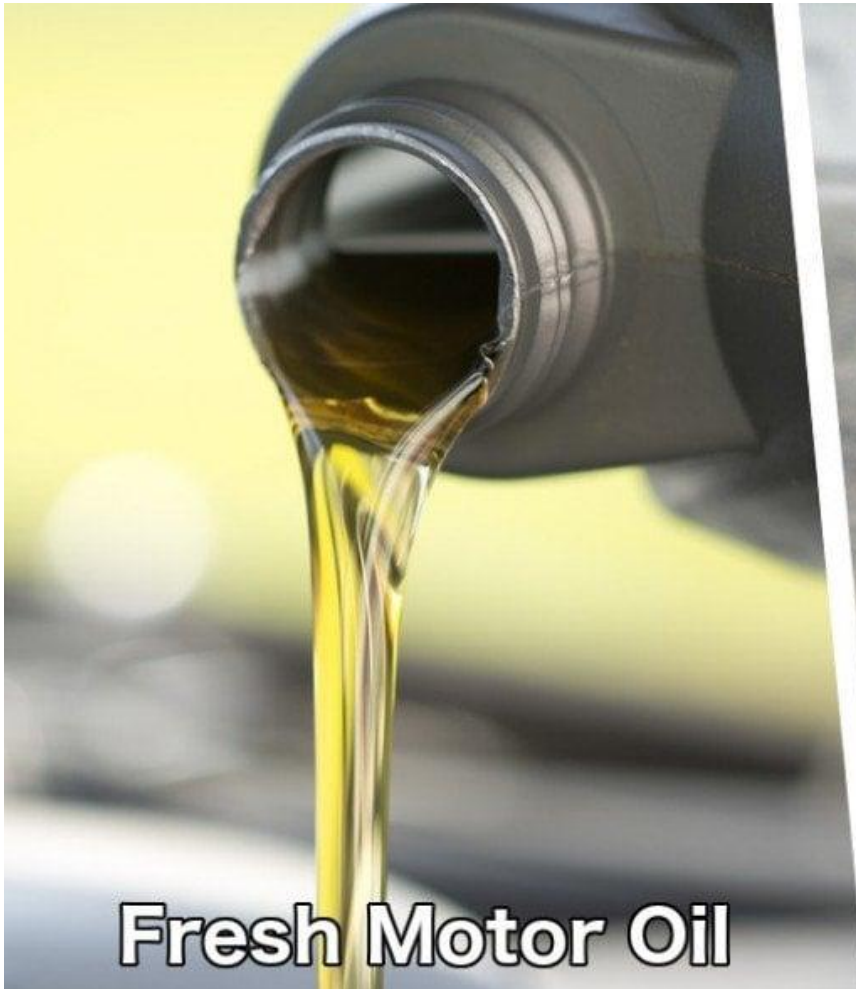
Forced feed lubrication system

Engine Lubrication System

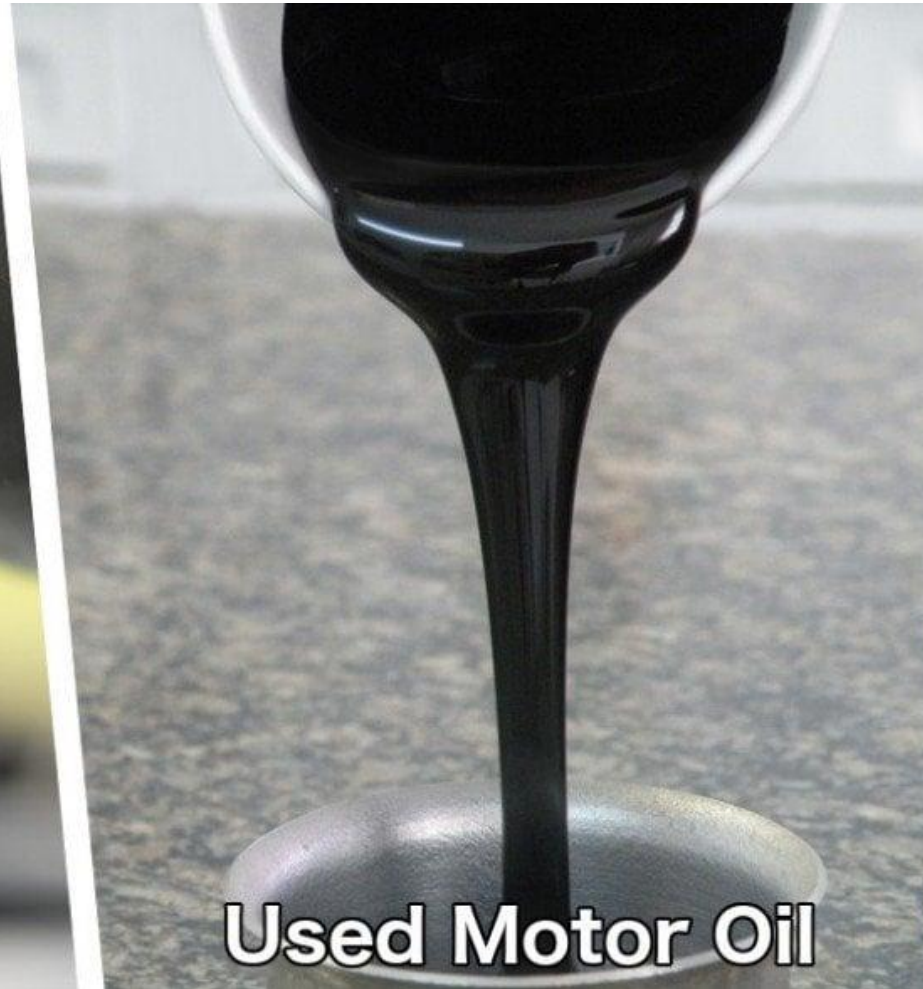


<https://www.youtube.com/watch?v=mmmcyj53TNic>

Engine Oil



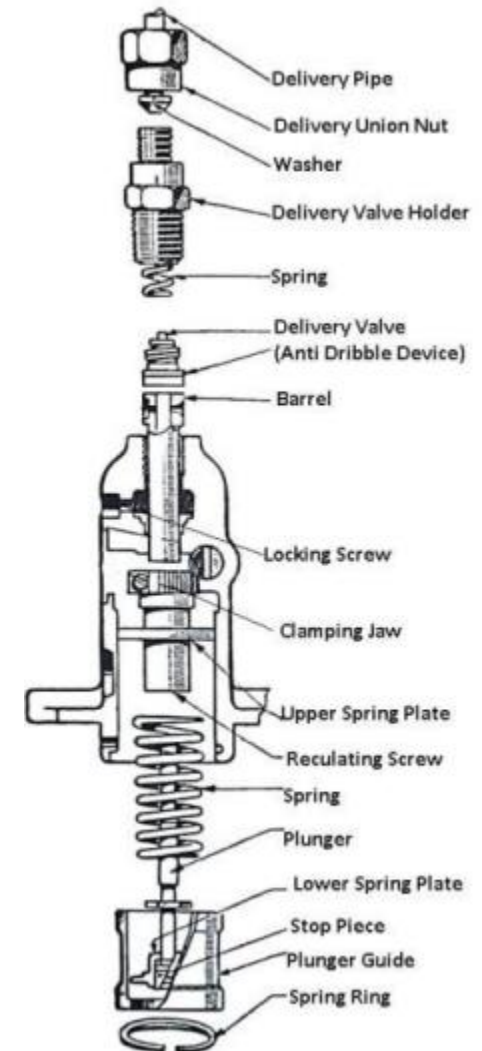
Fresh Motor Oil



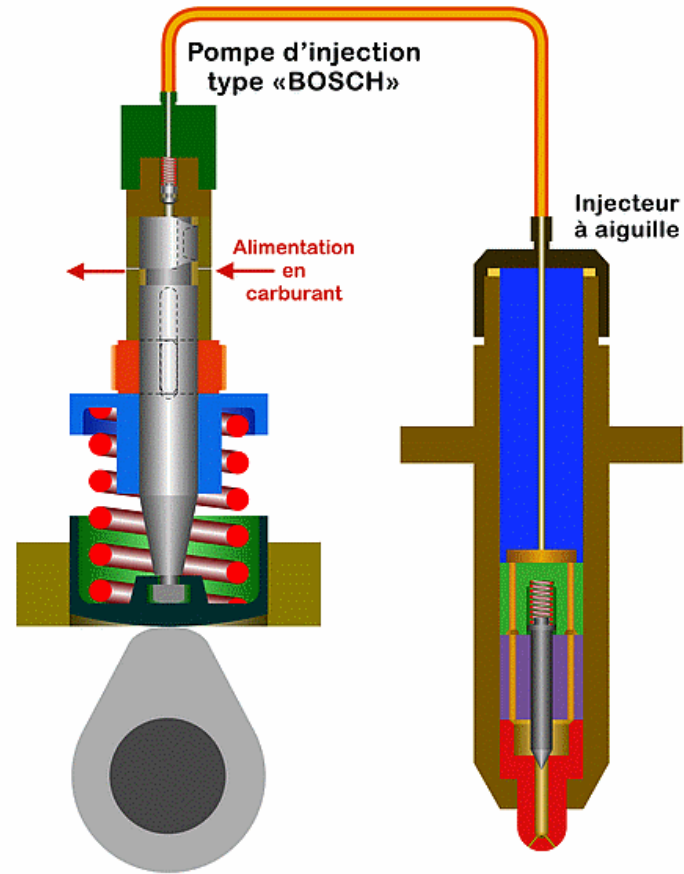
Used Motor Oil

3. Fuel Pump

- Purpose:- to meter the correct quantity of fuel and deliver it at the correct time to the engine cylinder according to the varying load and speed requirements



4. Fuel injection



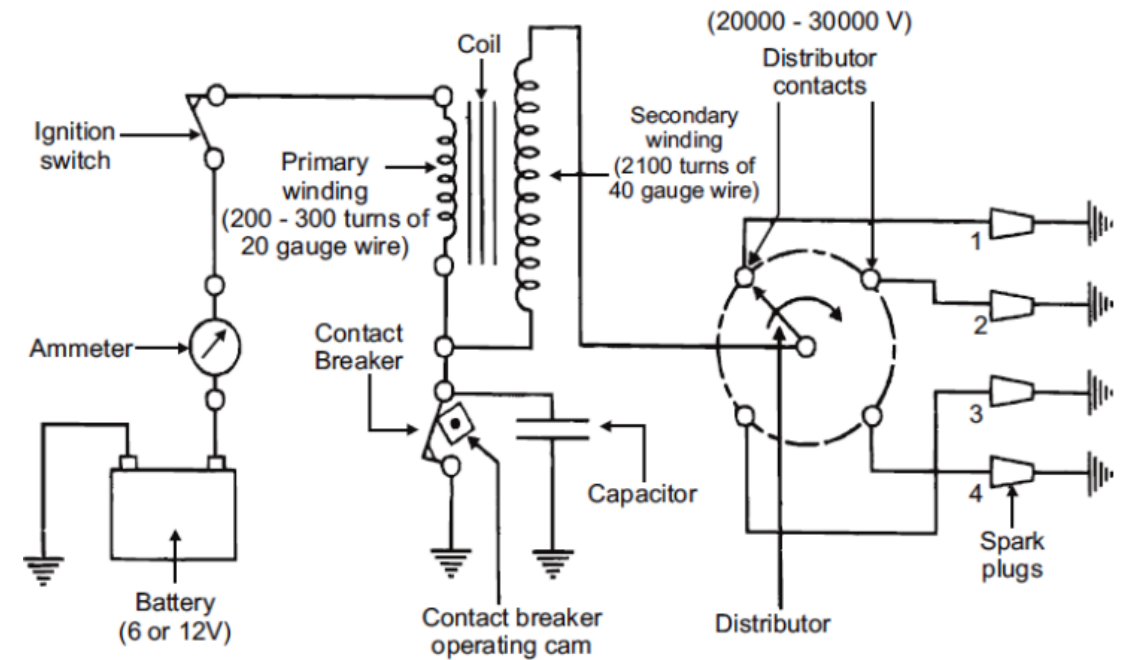
5. Ignition system

- Works on electromagnetic induction principle
- Two types
- Battery or Coil Ignition System, ex: 4 wheeler engines, 2 wheeler engines with battery
- Magneto Ignition System, ex: kick start engines



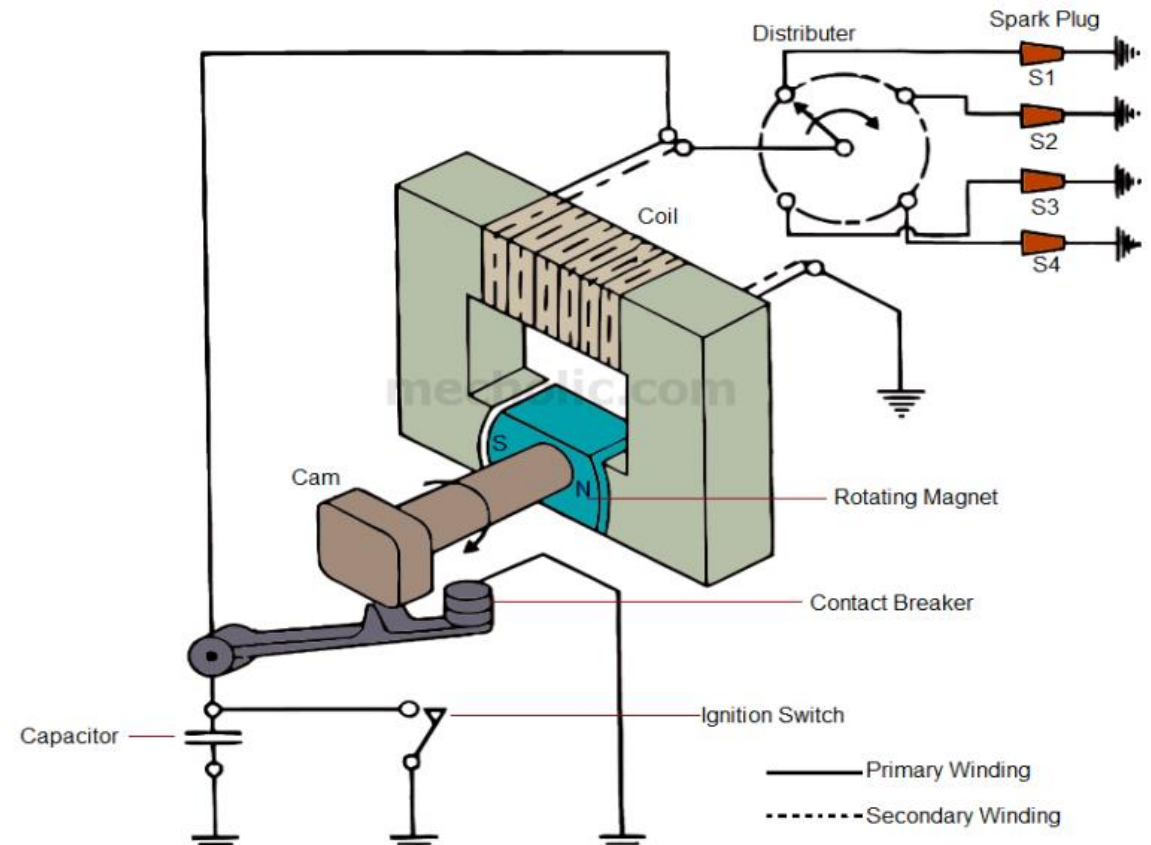
5a. Battery Ignition System

- Primary Circuit : It consists of 6 or 12 V battery, primary winding
- Secondary Circuit : It consists of secondary winding
- Low voltage current flows through primary windings,
- Secondary voltage goes upto 30000V
- High voltage current jumps the spark plug gap, it produces the spark

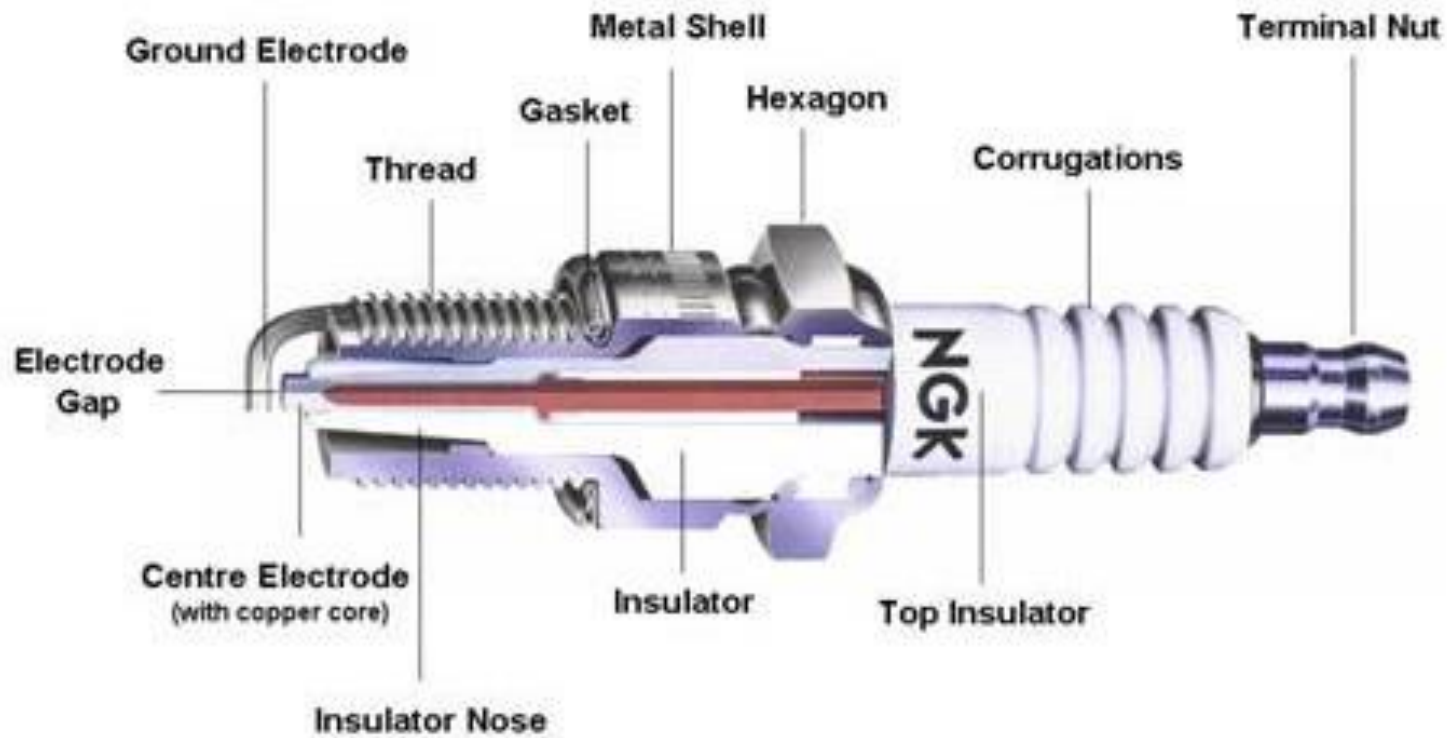


5b. Magneto Ignition System

- Magneto will produce and supply the required current to the primary winding.
- We can have rotating magneto with fixed coil or rotating coil with fixed magneto for producing and supplying current to primary, remaining arrangement is same as that of a battery ignition system.



Spark Plug

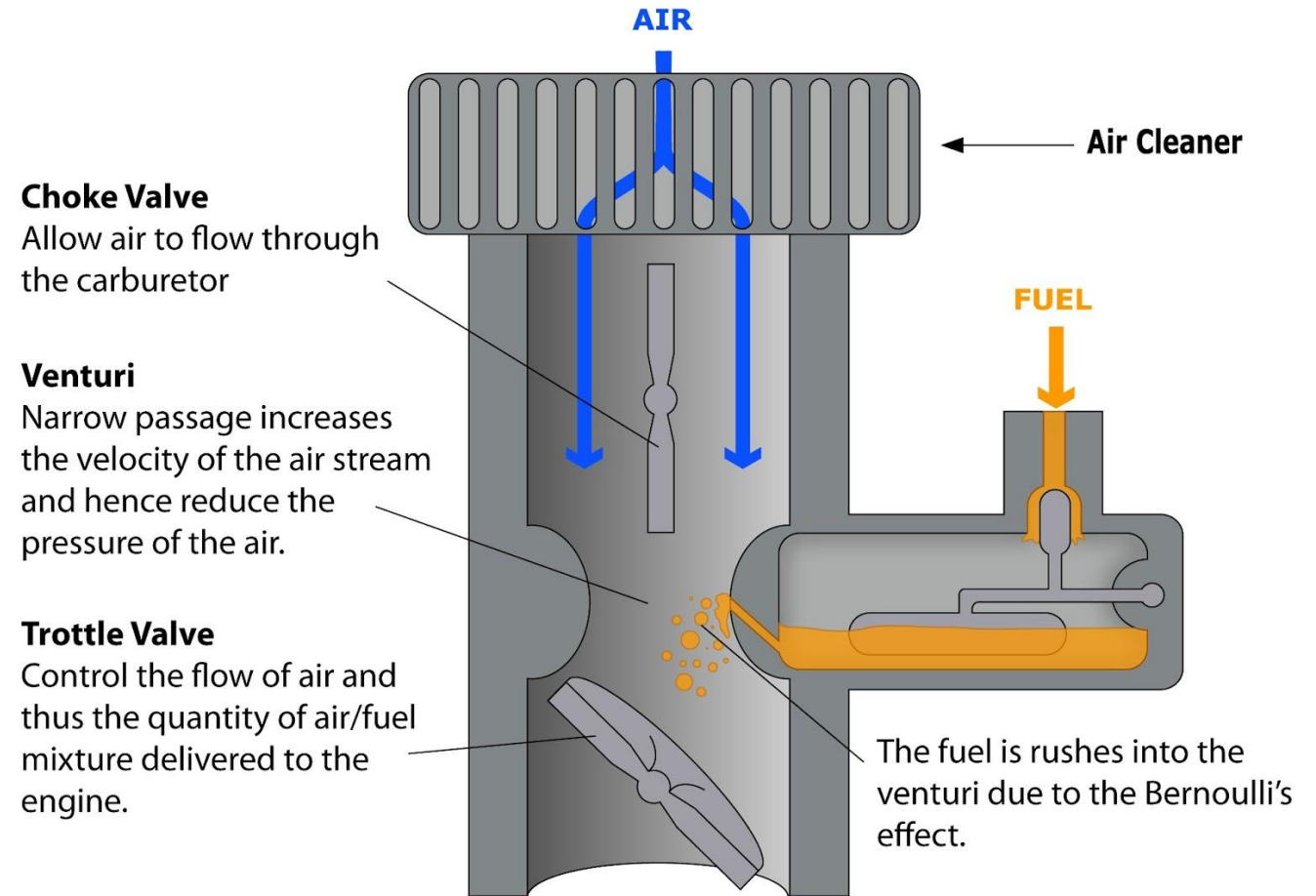


Used Vs New Spark Plug



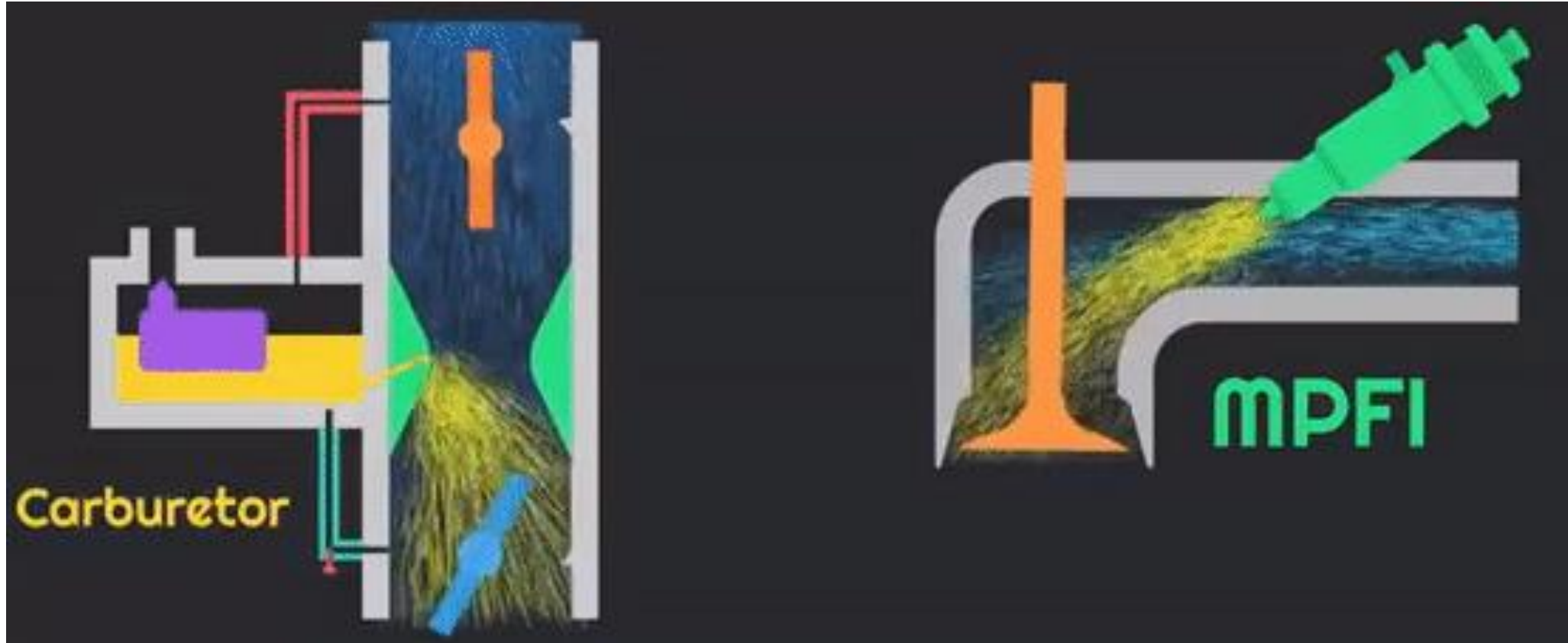
6. Carburetted and MPFI engine

- Carburettor:- Mix air with fuel
- Mix fuel with intake air when it crosses through venturi tube
- Working principle:-
Bernoulli's principle

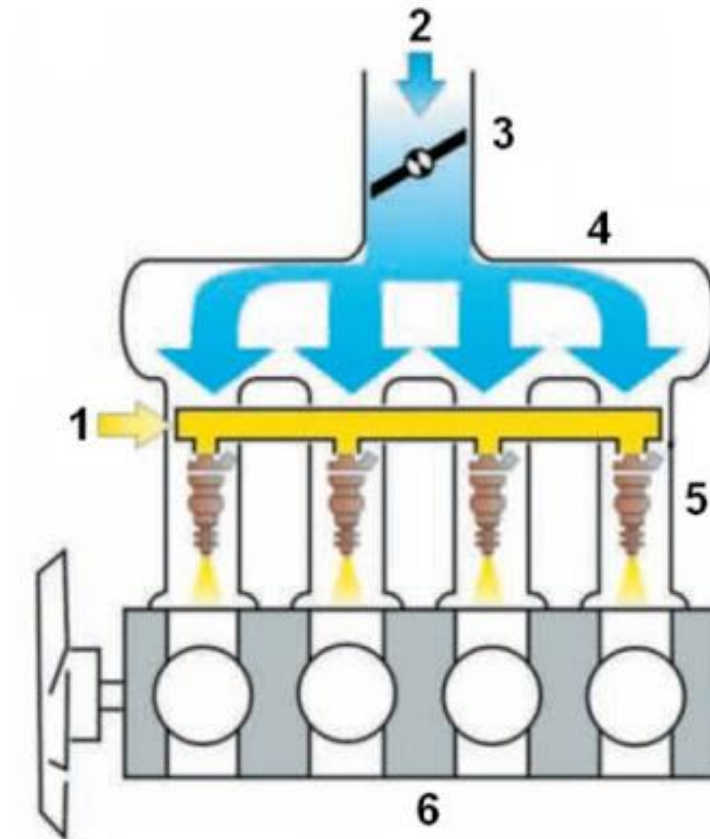


Basic Carburetor

Carburetor and MPFI



Carburetted and MPFI engines



[\[HINDI\] How MPFI System Works? Multi Point Fuel Injection System - YouTube](#)

Useful Links

[WORKING OF CARBURETTOR: How carburattor works? \(Animation\). – YouTube](https://www.youtube.com/watch?v=Ll62z9TJqCg&t=344s)
<https://www.youtube.com/watch?v=Ll62z9TJqCg&t=344s>



Thank You